Xiangling Xu (许湘灵)

My Website
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Google Scholar
Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*

2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00

2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Gravity with Noncommutative Geometry, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

1. L. J. Dmello, **X. Xu**, M.-O. Renou, and D. Gross, *An operational characterization of finite-dimensional quantum theory*, Oct. 2, 2026. arXiv: 2610.03694 [quant-ph]. URL: https://arxiv.org/abs/2610.03694.
2. **X. Xu**, I. Klep, and M.-O. Renou, *Quantum inflation and source-transfer hierarchies are complete for tree networks*, Sep. 20, 2026. arXiv: 2609.19310 [quant-ph]. URL: https://arxiv.org/abs/2609.19310.
3. F. M. Kalarde, S. Boreiri, **X. Xu**, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou, *Fermions are fundamentally more nonlocal than bosons*, Jun. 11, 2026. arXiv: 2606.12363 [quant-ph]. URL: https://arxiv.org/abs/2606.12363.
4. **X. Xu**, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou, *The bulk spectral gap is certifiable from above but uncomputable from below*, Jun. 2, 2026. arXiv: 2606.03836 [quant-ph]. URL: https://arxiv.org/abs/2606.03836.
5. F. M. Kalarde, **X. Xu**, and M.-O. Renou, *Comment on "quantum theory based on real numbers cannot be experimentally falsified": On the compatibility of physical principles with information theory for fermions*, Apr. 2026. arXiv: 2604.07425 [quant-ph]. URL: https://arxiv.org/abs/2604.07425.
6. M.-O. Renou, **X. Xu**, and L. T. Ligthart, "Two convergent npa-like hierarchies for the quantum bilocal scenario," *Journal of Mathematical Physics*, vol. 67, no. 1, p. 012 203, Jan. 2026, ISSN: 0022-2488. DOI: 10.1063/5.0211008. eprint: https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf.
7. M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, Contributed talk at QIP2026, 2025. arXiv: 2509.25145 [quant-ph]. URL: https://arxiv.org/abs/2509.25145.

8. I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, Contributed talk at QIP2026, 2025. arXiv: 2507.17006 [quant-ph]. URL: https://arxiv.org/abs/2507.17006.

9. **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. URL: https://arxiv.org/abs/2505.22309.

10. X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. DOI: https://doi.org/10.1093/mnras/stac3500.

11. **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023. DOI: https://doi.org/10.3929/ethz-b-000613670.

Talks & Conference Activities

Invited Talks

Jun 2026 — *2026 SIAM Conference on Optimization (OP26)*, Edinburgh, United Kingdom
Dec 2025 — *Quantum Meets*, online at https://quantum-meets.github.io
Jul 2025 — *IWOTA 2025 Twente*, Twente, Netherland
Oct 2023 — *Saclay Quantum Seminar*, Inria Saclay, France
Jun 2025 — *IQC-PCQT-Quantum Saclay Workshop*, Paris, France

Contributed Talks

Nov 2026 — *Journées Informatique Quantique 2026*, Toulouse, France
Aug 2026 — *QCrypt 2026*, Ottawa, Canada
Jan 2026 — *QIP 2026 Riga* – 2 contributed talks, Riga, Latvia
Nov 2025 — *3rd Colloquium GDR TeQ*, Grenoble, France
Oct 2025 — *YQIS25*, Barcelona, Spain
Sep 2024 — *Causalworlds 2024*, Waterloo, Canada
Nov 2023 — *Workshop Defi EQIP*, Lyon, France

Talks during group visits

Sep 2026 — *Quriosity*, Inria & Télécom Paris & IP Paris
Aug 2026 — *Ranyiliu Chen*, Quantum Science Center of Huangdong-HongKong-Macao Greater Bay Area
Jul 2026 — *Penghui Yao*, Nanjing University
Feb 2026 — *Jean-Daniel Bancal*, CEA, Institut de physique théorique
Dec 2025 — *Jie Wang*, AMSS-CAS, Beijing
Nov 2025 — *Thomas Vidick*, EPFL
Oct 2025 — *Jens Eisert*, Freie Universität Berlin
Feb 2025 — *QAT Team*, ENS Paris
Jul 2024 — *William Slofstra*, University of Waterloo
Dec 2023 — *David Gross*, University of Cologne

Poster Sessions

Aug 2026 — *CQCC & GQSF 2026*, Shenzhen, China

Talks & Conference Activities (continued)

Nov 2024
- *2nd Colloquium GDR TeQ*, Paris, France
- *YQIS24*, Paris, France

Other Participation

Sep 2026
- *Quantum Information Aussois*, Aussois, France

Mar 2025
- *828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory*, Bad Honnef, Germany

Sep 2024
- *VQF Vienna Quantum Foundations Conference*, Vienna, Austria

Nov 2023
- *POP23 Workshop on Future Trends in Polynomial OPtimization*, Toulouse, France

Collaborations

Invited Group Visits

Oct-Nov 2026
- *David Gross*, University of Cologne
  Invited three-week visit

Jul 2026
- *Penghui Yao*, Nanjing University
  Invited one-week visit

Dec 2025 & May 2026
- *Jie Wang*, AMSS-CAS, Beijing
  Two invited one-week visits

Oct-Nov 2025
- *Thomas Vidick*, EPFL
  Invited two-week visit

Sep 2024
- *Igor Klep*, University of Ljubljana
  Invited two-week visit funded by Erasmus+ Mobility program

Jul 2024
- *William Slofstra*, Institute for Quantum Computing, University of Waterloo
  Invited two-week visit

Dec 2023
- *David Gross*, University of Cologne
  Invited one-week visit

Organizations

Mar 2025 - Mar 2026
- Coorganizer for *Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory*
  Coordinating the Gaspard Monge Visiting Professor program for Igor Klep
  Co-planning the seminar and the schedule for invited speakers

Mar 2025
- Coorganizer for *828. WE-Heraeus-Seminar at Bad Honnef*
  Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts
  Coordinating with the organizers and inviting some of the speakers

Feb 2024 - Now
- Coorganizer of *Project COMPUTE for Quantera*
  Coordinating with the members, project poster and website

Sep 2023
- Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*
  Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

2026 — *Erasmus+ Mobility Program*, for visiting the University of Cologne
2024 — *Erasmus+ Mobility Program*, for visiting the University of Ljubljana
2020 — *McNab Undergraduate Scholarship*, University of Toronto
2017 - 2020 — *Dean's List Scholar*, University of Toronto
2018 — *The Class of 3To and Associates Scholarship in Mathematics and Physics*, University of Toronto
International Exchange Award, University of Toronto
Bronze Medal, 2017 University Physics Competition
2017 — *Eileen Doman Award*, University of Toronto
Summer Undergraduate Research Program, Canadian Institute for Theoretical Astrophysics
2016 — *President's Scholar of Excellence Program*, University of Toronto

Teaching Experience

Lecture

Dec 2025 — *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay

Nov 2024 — *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay

Teaching Assistance

2022-2023 — *Linear Algebra I & II*
Department of Mathematics, ETH Zürich

2021-2022 — *Analysis I & II*
Department of Mathematics, ETH Zürich

Skills

Languages — Chinese (native), English (fluent), German (beginner)
Coding — Python (Ncpol2sdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, LaTeX

Referees

Available on Request